Defense-Aware Attacks on Content Disarm and Reconstruction and JPEG as Defenses for a Vision-Language Model

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Abstract

Content Disarm and Reconstruction (CDR) is recommended as a defense against adversarial images and prompt injection for multimodal models, but the evidence for it comes from attackers who do not know the defense exists. We attack a vision-language model (LLaVA-1.5-7B) with a targeted L_∞ attack that is optimized through the defense (backward pass through a differentiable surrogate, forward pass through the real defense where it can run) and score every result through the real tool. The defenses are JPEG quality 50 and Dangerzone, an open-source CDR tool, run with an adapter that does not add a JPEG stage of its own. On 200 images with a pre-registered analysis of six paired tests, the defense-aware attack beats the defense-unaware attack in all six. At 16/255 it succeeds on 21% of the images through JPEG 50 and on 76.5% through Dangerzone, against 0% and 1.5% for the unaware attack. At 8/255 the rates are 9% and 29%. Under defense-aware attack Dangerzone is easier to break than JPEG 50, by 20 points at 8/255 and 55.5 points at 16/255. All attacks stopped at 500 steps before their loss had flattened, so every rate is a lower bound, and the Dangerzone and JPEG attacks are not guaranteed to be equally strong. The data support the claim that neither defense holds against an attacker who knows it. They give no support for the claim that CDR protects better than JPEG. They cover one model, one benign target and one CDR tool.

1 Introduction

A small, bounded change to the pixels of an image can steer what a vision-language model (VLM) says about it [1]. A common model-agnostic defense is to re-encode the input before the model sees it [2]. Security practice adds another candidate: Content Disarm and Reconstruction (CDR) parses a file, keeps only content it can validate, and rebuilds a clean file. Published guidance already recommends multimedia CDR against prompt injection in agent pipelines [3].

Whether CDR defends a model against adversarial pixels is an empirical question, and the answer depends on the attacker. A defense that works against an attacker who ignores it often fails once the attacker optimizes through it [4, 5]. In an earlier non-adaptive study with the same model and images, every defense we tried removed almost all of a targeted attack: of 185 attacks that worked without a defense, none survived Dangerzone or JPEG 50. That result says little about an attacker who adapts, and it has a second problem described in Section 2.5: the Dangerzone adapter we used embedded a JPEG 75 image, so its result could not be credited to Dangerzone.

This paper reports the defense-aware follow-up. It asks two questions:

- **H1.** Does an attacker who optimizes through the defense recover success that a defense-unaware attacker does not have?
- **H2.** Under defense-aware attack, does Dangerzone differ from JPEG 50?

Both were registered before any adaptive result existed (Section 2.6). CDR is an existing technique and Dangerzone [6] is an existing tool. The contribution is the measurement: paired, pre-registered, with the real defense in the scoring path, and with the limits of the attack stated next to the numbers.

2 Setup

2.1 Model, data and success criterion

The model is LLaVA-1.5-7B [7] (`llava-hf/llava-1.5-7b-hf`) with its CLIP ViT-L/14-336 vision tower [8], loaded in 4-bit NF4 with fp16 compute and decoded greedily. The question is "What is the main object in this image? Answer with a single lowercase word."

The 200 images are 10 from each of 20 ImageNet classes [9], resized to 336×336. The set is fixed, and the 20 classes are not a random draw from 1000. Each image has a benign target label, the next class in sorted order. An attack succeeds when the model's greedy answer through the defense contains the target word. This measures a targeted output hijack to an innocuous label. No harmful content is generated.

2.2 Defenses

Every number in this paper is scored through the real defense.

- **JPEG 90, 75, 50:** Pillow, 4:2:0 subsampling.
- **Dangerzone 0.11.0** [6], run through rootless Podman. It works on documents, so each image is wrapped in a one-page PDF, sanitized, and rasterized back to 336×336. We use the *lossless adapter*, which embeds the image in the PDF as lossless bytes. The *legacy adapter* uses Pillow's PDF writer, which re-encodes the image as JPEG 75 first. Section 2.5 explains why the two are kept apart.

ICDR [10, 11] was planned as a second CDR tool. Its committed code depends on a commercial imaging library and does not compile, so it was not run.

2.3 Defense-aware attack

The attack is targeted L_∞ projected gradient descent [12] with a sign-gradient step of 1/255, a random start, seed 1234, and 500 steps. The loss is teacher-forced cross-entropy toward the target word, computed on the image *after* the defense. Gradients come from a backward pass through a differentiable surrogate with a straight-through estimator for rounding (BPDA [4]). The two surrogates are:

- **JPEG 50.** A differentiable JPEG (blockwise DCT, quantization with the libjpeg tables, straight-through rounding). The forward pass is the real Pillow codec, so the model is attacked on exactly the pixels it will see.
- **Dangerzone.** The real tool cannot run on the GPU host, so a separable banded linear map (bandwidth 8) is fitted to the real tool's input and output pairs (600 training and 200 held-out pairs). Its held-out PSNR against the real output is 48.9 dB on clean and 48.1 dB on adversarial inputs. The attack uses the surrogate in both passes. Final success is always scored through the real tool.

The update has no momentum and a constant step, so the iterate is the whole optimizer state and an interrupted attack resumes exactly. Budgets are $\epsilon \in \{8, 16\}/255$. A change of 16/255 lets each colour channel move by at most 16 of 255 levels, about 6%.

2.4 Non-adaptive baseline

The defense-unaware condition is the stored attack from the earlier study: PGD against the undefended model, 200 steps of 1/255, same images, targets and seed, scored through each real defense. The stored arrays were rescored in a control run (`controls_v1`), which is the baseline here. Without a defense this attack reaches 10.5%, 26.5% and 56% of the images at 4, 8 and 16/255 (21, 53 and 112 of 200).

2.5 The adapter confound

The legacy Dangerzone adapter produces a JPEG 75 image, then a 150 dpi raster, then Dangerzone's own conversion. The embedded image is bit-identical to Pillow's JPEG 75 output. JPEG 75 alone defeated 184 of the 185 attacks that worked undefended, so the earlier Dan-

Table 1: Targeted success through the real defense, $N = 200$ per cell. Rates have exact 95% intervals. The difference is adaptive minus non-adaptive with a Newcombe interval. b/c are the discordant pairs (success only under the adaptive attack / only under the non-adaptive attack). p_{Holm} is over the six registered tests.

Defense	ϵ	Non-adaptive	Adaptive	Diff. (pts)	b/c	p_{Holm}
JPEG 50	8/255	0/200 (0%)	18/200 (9%)	9.0 [5.3, 13.8]	18/0	8×10^{-6}
Dangerzone	8/255	1/200 (0.5%)	58/200 (29%)	28.5 [22.5, 35.1]	57/0	6×10^{-17}
JPEG 50	16/255	0/200 (0%)	42/200 (21%)	21.0 [15.6, 27.2]	42/0	1×10^{-12}
Dangerzone	16/255	3/200 (1.5%)	153/200 (76.5%)	75.0 [68.2, 80.4]	150/0	8×10^{-45}

gerzone result could not be attributed to Dangerzone. The lossless adapter was added for this study. The two adapters are separate conditions in every table, and no number pools them. Clean-image preservation is similar for both: the model's own clean answer is unchanged on 93.5% of images through the lossless adapter and 91.5% through the legacy one (JPEG 50: 92%).

2.6 Pre-registration, tests and statistics

The analysis was registered in the repository before any adaptive result existed (files `phase2_preregistration.md` and `analysis_plan.yaml`). It is an internal registration timestamped by the commit history, not an external registry entry. The confirmatory family has six tests: H1 for JPEG 50 and for Dangerzone, and H2, each at 8/255 and 16/255. Each test pairs the same 200 images across two attack conditions. We report exact McNemar tests on the discordant pairs, Clopper–Pearson intervals for rates, Newcombe intervals for paired differences, and Holm correction over the six tests at $\alpha = 0.05$. The minimum effect of interest is 15 points. At $N = 200$ and a discordant rate of 0.3 the simulated power to detect a 15-point difference is 0.89, and differences below about 10 points cannot be resolved. A 50-image screen preceded the confirmatory runs. It was exploratory, and the four confirmatory conditions were chosen after it (Appendix A).

2.7 Compute

The attacks take 12 to 15 minutes per image on a Colab T4, about 180 GPU hours for the four confirmatory runs. State lived on Drive, and each session resumed from the stored iterate. Sanitizing ran on a local machine with rootless Podman. GPU runs are not bit-reproducible.

3 Results

3.1 H1: knowing the defense recovers success

Table 1 and Figure 1 give the targeted success rates. All four adaptive differences are significant after Holm correction, and in every case the discordant pairs run in one direction only: 42 images flip to success through JPEG 50 and none flip the other way at 16/255, and 150 against 0 through Dangerzone.

The JPEG 50 effect at 8/255 is the smallest, 9 points with an interval of 5 to 14. Its interval lies below the 15-point minimum effect of interest, so it is a detected effect that is smaller than the one the study was powered to resolve.

3.2 H2: Dangerzone against JPEG 50 under defense-aware attack

With each defense attacked through its own surrogate for the same 500 steps, the success rate is higher against Dangerzone than against JPEG 50: 29% against 9% at 8/255 (difference 20 points, interval 14 to 26, 43 discordant pairs against 3) and 76.5% against 21% at 16/255 (55.5 points, interval 48 to 62, 113 against 2). Both are significant after correction, and both intervals exclude the 15-point effect. Figure 2 shows all six differences.

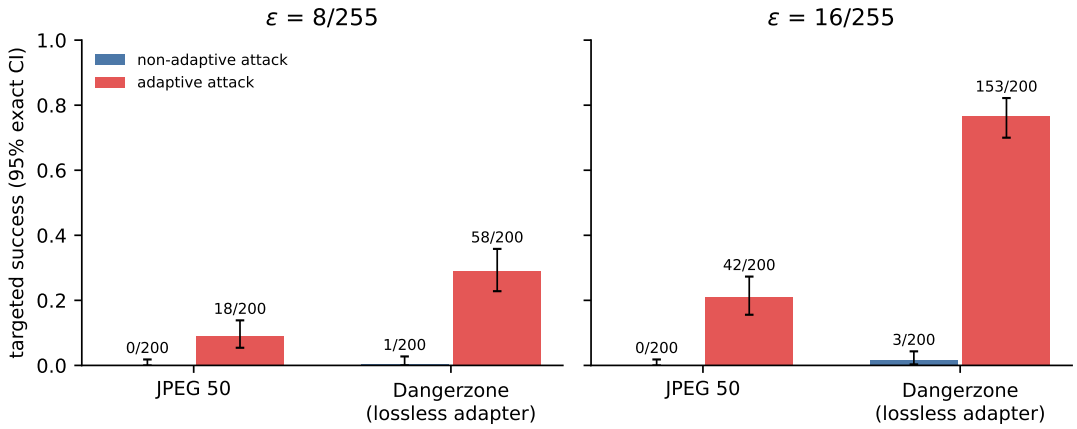


Figure 1: Targeted success through the real defense for the non-adaptive and the adaptive attack (error bars: exact 95% intervals; counts above the bars).

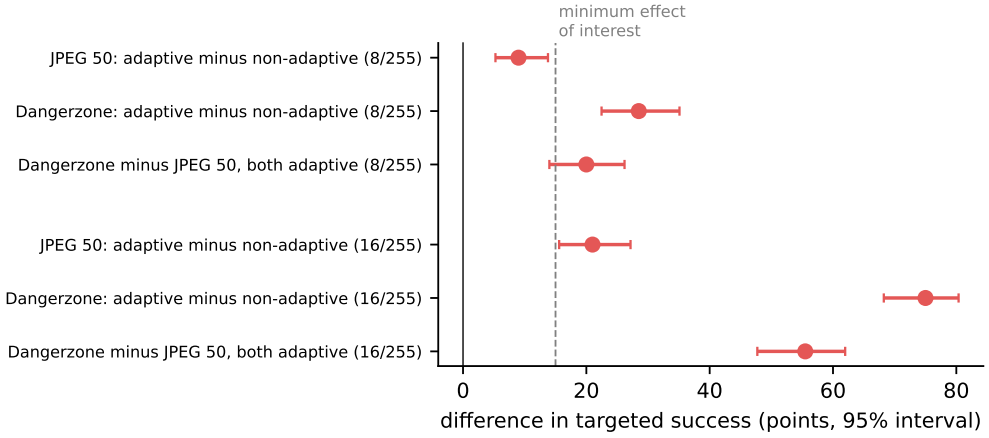


Figure 2: The six registered paired differences with 95% intervals. The dashed line is the 15-point minimum effect of interest.

This is a comparison of two attacks as well as two defenses, and it should not be read as a ranking of the defenses (Section 5).

3.3 The adaptive images are specific to their defense

Scored through other defenses, the images optimized through Dangerzone almost never succeed: at 16/255 they reach 1, 0 and 0 of 200 through JPEG 90, 75 and 50 (16 of 200 undefended). The images optimized through JPEG 50 do not carry over to its neighbours either: at 16/255 they succeed on 12 and 10 of 200 through JPEG 90 and 75, about the 11 they reach undefended, and 42 through JPEG 50. At 8/255 the counts are 12 and 3, against 18 through JPEG 50. The attack therefore exploits the specific defense it was given, and an attacker who wants success through a deployed defense has to attack that one.

3.4 Optimization and validity gates

Figure 3 shows the median target loss during the attack. The defended loss (measured on the real defense after the attack) is lower for the adaptive image than for the non-adaptive image on 200 of 200 images in every run (medians 0.43 against 5.0 for Dangerzone at 16/255, 1.63 against 5.7 at 8/255; 2.70 against 7.5 and 4.46 against 7.7 for JPEG 50). Every run passed the

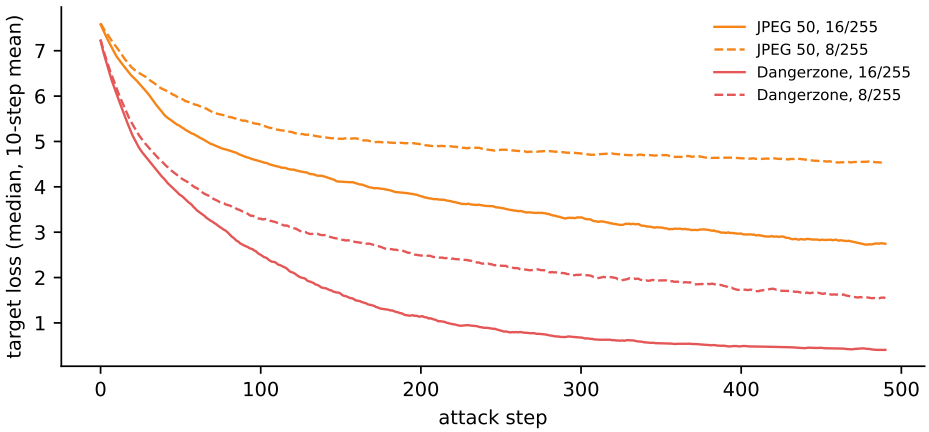


Figure 3: Median target loss during the 500-step attack (10-step mean). Dangerzone curves use the surrogate forward pass, JPEG curves the real codec.

gates registered before the study: all jobs finished, the perturbation stayed inside the ball, and no real-defense evaluation failed. The Dangerzone surrogate meets the 40 dB fidelity threshold.

None of the attacks converged. The median loss still fell by 17.7% and 13.6% over the last 100 steps, relative to the 100 before them, of the Dangerzone runs (16 and 8/255) and by 7.8% and 1.9% for JPEG 50. The JPEG 50 attack ends at a loss of 2.7 to 4.5, well above the 0.4 to 1.6 that the Dangerzone attack reaches, so it is farther from success at the same step count. In a pilot on ten images at 16/255, the JPEG 50 success rate rose with steps (0/10 at 200, 4/10 at 500, 6/10 at 1000, 7/10 at 2000). Every rate in Table 1 is therefore at or below what a longer attack would give.

3.5 Exploratory screen: the legacy adapter

A 50-image screen preceded the confirmatory runs and included the legacy adapter, which was not repeated at $N = 200$. At 16/255 the success rate through real Dangerzone was 42% with the legacy adapter and 80% with the lossless adapter (JPEG 50: 24%). At 8/255 the rates were 14%, 26% and 12%. These are exploratory, with Holm correction inside the screen's own family. The legacy adapter was harder to attack than the lossless one. Its extra JPEG 75 stage is the likely reason, which would make the difference an effect of JPEG and not of CDR, but we did not test this directly.

4 Discussion

An attacker who knows the defense gets large gains against both tools. Against Dangerzone with the lossless adapter the gain is 75 points at 16/255, and JPEG 50 is also broken on 21% of the images at the same budget. The earlier result that every defense removes almost all of the attack describes an attacker who ignores the defense. It should not be cited as a statement about robustness.

Dangerzone's sanitization removes what a defense-unaware perturbation depends on, and JPEG 50 does too. Neither removes the information an attacker needs once the defense is in the optimization loop. The lossless Dangerzone pipeline is close to a resampling and a re-rasterization, which are smooth and nearly invertible at the scale of the perturbation, so gradients pass through them well. JPEG 50 quantizes in the DCT domain, which discards more information, and the attack makes less progress through it. We did not run an experiment that isolates this explanation, and it should be treated as a hypothesis.

The data do not support a claim that CDR protects better than JPEG, in either direction. The H2 result favours JPEG 50 at equal steps, but the JPEG attack stops at a higher loss

and both surrogates differ in kind (real forward for JPEG, fitted forward for Dangerzone). A converged attack on JPEG 50 could erase the gap or reverse it.

5 Limitations

- **Unconverged attacks.** All rates are lower bounds. A zero under these attacks would not have shown robustness, and the nonzero rates understate what a longer attack gets.
- **Step budgets differ between the two attack conditions.** The adaptive attack runs 500 steps and the stored non-adaptive attack 200. We did not run a 500-step non-adaptive control, so H1 combines knowledge of the defense with a larger step budget, and we cannot say how much of each gain comes from the extra steps. The size of the gains and the pilot's rise with steps both suggest that knowledge of the defense is the larger part, but this is a judgment and not a measurement.
- **Unequal attack strength across defenses.** JPEG 50 is attacked through the real codec and a straight-through gradient, Dangerzone through a fitted linear surrogate. H2 compares both the defenses and the attacks.
- **One model, one target rule, 20 classes, one CDR tool.** We did not run a second VLM, a typographic or prompt-injection attack, ICDR, MetaDefender Core or other CDR products, or 4/255 in the adaptive setting. Results do not carry over to those cases.
- **The surrogate is an approximation.** Its 48 to 49 dB fidelity is high, and success is scored on the real tool, but the attack direction is only as good as the surrogate's Jacobian.
- **One seed and a stored baseline.** GPU runs are not bit-reproducible. Thirty-nine of the 600 non-adaptive rows in the original run used pixels different from the stored arrays. This changes the undefended count by one image and none of the defended counts (Appendix A). The baseline here comes from the rescored stored arrays.
- **The registration is internal.** It is timestamped by the commit history and was never filed with an external registry.

6 Conclusion

Against an attacker who optimizes through the defense, both JPEG 50 and Dangerzone fail on a substantial share of 200 images at 16/255 (21% and 76.5%) and a smaller share at 8/255 (9% and 29%), where the defense-unaware attack reached 0% and 0.5%. The study supports the statement that defenses evaluated against defense-unaware attacks should not be called robust, and it does not support a ranking of CDR against JPEG. Closing the gaps listed above, starting with a longer attack and a 500-step non-adaptive control, is the direct next step.

Reproducibility

Code, configurations, the registration and the analysis scripts are in the project repository. The pre-registered analysis is `scripts/analyze_confirmatory.py` under `configs/analysis_plan.yaml`, and the figures come from `scripts/make_paper_figures.py`. The four confirmatory runs are `conf_jpeg50_e08`, `conf_jpeg50_e16`, `conf_dz_ll_e08` and `conf_dz_ll_e16`, with the non-adaptive baseline in `controls_v1`. The non-adaptive baseline tag is `baseline-nonadaptive-v1`. The model weights and the ImageNet images come from their public sources.

A Deviations from the registration

- The registered baseline-reproduction gate G5 (at least 99% raw-answer agreement between the rescored controls and the original run) failed at 96.8% (581 of 600) and is reported as failed. The 19 differing answers all came from the 39 rows whose input pixels differed between the two runs, most likely because those attacks were crafted again after the first evaluation. Two diagnostic gates, G5a and G5b, were added after seeing this. Both restrict

to identical inputs and pass (561 of 561). The undefended count changes by one image; the JPEG results are identical in both runs. The earlier report needs an erratum, and its conclusions do not change.

- The registered JPEG 50 extension schedule was changed from 400 to 500, 1000 and 2000 steps after the 200-step pilot loss was still falling.
- The screen images were stratified over the 20 classes instead of taking the first 50 in manifest order, which would have covered only 5 classes. 4/255 was left out of the adaptive runs for compute reasons.
- The confirmatory step budget was set to 500 for all adaptive attacks, a compute cap chosen after the screen, so the attacks are at equal steps but unconverged.
- The confirmatory conditions (JPEG 50 and lossless Dangerzone at two budgets) were chosen after the screen. The legacy adapter was not run at $N = 200$.

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